

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCVC304 - HYDRAULICS-I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following fluid properties is dimensionless?
 - A) Density
 - B) Specific gravity
 - C) Specific weight
 - D) Viscosity
- 2 In a U-tube differential manometer, if the manometric liquid is mercury and the fluid in both limbs is water, the pressure difference measured will be:
 - A) Equal to zero
 - B) In terms of mercury head
 - C) In terms of water head
 - D) Negative pressure
- 3 For a floating body, if the metacentric height is negative, the body:
 - A) Is in stable equilibrium
 - B) Will sink
 - C) Is in unstable equilibrium
 - D) Is in neutral equilibrium
- 4 The continuity equation is based on the principle of:
 - A) Conservation of energy
 - B) Conservation of momentum
 - C) Conservation of mass
 - D) Conservation of velocity
- 5 In an ideal fluid flow, Bernoulli's equation assumes:
 - A) Viscous and compressible flow
 - B) Viscous and incompressible flow
 - C) Incompressible and inviscid flow
 - D) Compressible and turbulent flow
- 6 In a flow net, flow lines and equipotential lines always intersect at:
 - A) 30°
 - B) 45°
 - C) 60°
 - D) 90°

- 7 For fully developed laminar flow between two parallel plates (one moving), the velocity profile is:
 A) Linear
 B) Logarithmic
 C) Parabolic
 D) Asymmetric
- 8 The Reynolds number for transition from laminar to turbulent flow in a pipe is approximately:
 A) 500 B) 1000 C) 2300 D) 4000
- 9 Which of the following is not a dimensionless number?
 A) Reynolds number
 B) Froude number
 C) Weber number
 D) Poiseuille number
- 10 Buckingham π -theorem helps in:
 A) Designing experiments using trial-and-error
 B) Reducing variables in dimensional analysis
 C) Creating time-series data
 D) Finding turbulent energy
- 11 A siphon works due to:
 A) Bernoulli's principle and gravity
 B) Viscous action
 C) Pumping force
 D) Surface tension
- Major loss in pipe flow is due to:
 A) Bends and fittings
 B) Entry and exit losses
 C) Friction in the length of the pipe
 D) Sudden contraction

Q2. Solve the following.

- A) (i) Develop the expression for the relation between gauge pressure P inside a droplet of liquid and the surface tension. 6
 (ii) Explain the following: Newtonian and Non-Newtonian fluids, vapour pressure, and compressibility.
- B) Prove that the total pressure exerted by a static liquid on an inclined plane submerged surface is the same as the force exerted on a vertical plane surface as long as the depth of the centre of gravity of the surface is unaltered. 6

Q3. Solve the following.

- A) The velocity components in a two-dimensional flow field for an incompressible fluid are expressed as 6

$$u = 2x + \frac{y^3}{3} - yx^2 ; v = -2y - \frac{x^3}{3} + xy^2$$

- (a) Obtain an expression for stream function Ψ
 (b) Obtain an expression for velocity potential Φ
- B) Derive Euler's equation of motion. Explain how this is integrated to get Bernoulli's equation. 6

Q4. Solve Any Two of the following.

- A) Derive the dynamic equation for gradually varied flow with assumptions in the form: 6

$$\frac{dy}{dx} = \frac{S_0 - S_f}{1 - F_r^2}$$

- B) What do you understand by 'Economical section' or 'Hydraulically most efficient section'? Derive condition for hydraulically most efficient rectangular section. 6

- C) With usual notation in hydraulic jump, prove the following relation for hydraulic jump in rectangular channel 6

$$\frac{y_2}{y_1} = \frac{1}{2} \left[-1 + \sqrt{1 + 8F_{r1}^2} \right]$$

Q5. Solve Any Two of the following.

- A) Define priming. Why it is necessary for a centrifugal pump? 6
- B) Enlist types of turbines. Explain with neat sketch Pelton wheel turbine. 6
- C) The internal and external diameters of the impeller of a centrifugal pump are 200 mm and 400 mm respectively. The pump is running at 1200 rpm. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially, and the velocity of flow is constant. Determine the work done by the impeller per unit weight of water. 6

Q6. Solve Any Two of the following.

- A) Explain methods of dimensional analysis. 6
- B) Explain different types of similarities that must exist between a prototype and its model. 6
- C) Explain the terms: distorted models and undistorted models. What is the use of distorted models? 6

*** End ***